One high-severity case, two wording rules

Case hlab1502_carba_pos · GPT-5.2, replicate 1. Only the knowledge supplied and the decision agent differ.

Identical input to all five configurations

```
Gene: HLA-B*15:02        Genotype: HLA-B typing positive for *15:02        Target drug: carbamazepine
Authored reference: carbamazepine: AVOID (potentially LETHAL — Stevens-Johnson syndrome / TEN risk)
```

A single configuration-blind function scores every response.

1. Free generation

Both wording rules pass

knowledge: none decision by: model

Locus failures: original 0/48; widened 0/48

```
DIPLOTYPE: HLA-B*15:02 positive (carrier; other allele not specified)
DRUG: Avoid carbamazepine; use an alternative (only consider if no options and benefits outweigh risks
           with informed consent and close monitoring)
```

2. RAG generation

Original fail; widened pass

knowledge: retrieved guideline prose decision by: model

Locus failures: original 45/48; widened 0/48

```
DIPLOTYPE: HLA-B*15:02/*X (at least one HLA-B*15:02 allele; second allele not reported)
DRUG: If carbamazepine-naive, do not use carbamazepine; use an alternative agent (if previously
           tolerated carbamazepine >3 months without cutaneous reaction, may cautiously consider use)
```

3. RAG-assisted execution

Both wording rules pass

knowledge: retrieved prose (call only) decision by: code

Locus failures: original 0/48; widened 0/48

```
DIPLOTYPE: HLA-B*15:02 positive

code applies the authored mapping  ->  carbamazepine: AVOID (potentially LETHAL — Stevens-Johnson
      syndrome / TEN risk)
```

4. Authored-rule generation

Both wording rules pass

knowledge: authored rule table decision by: model

Locus failures: original 0/48; widened 0/48

```
DIPLOTYPE: HLA-B*15:02 positive
DRUG: carbamazepine: AVOID (potentially LETHAL — Stevens-Johnson syndrome / TEN risk)
```

5. Authored-rule execution

Both wording rules pass

knowledge: authored rule table decision by: code

Locus failures: original 0/48; widened 0/48

```
DIPLOTYPE: HLA-B*15:02 positive

code applies the authored mapping  ->  carbamazepine: AVOID (potentially LETHAL — Stevens-Johnson
      syndrome / TEN risk)
```

Responses are reproduced verbatim from the deposited run records (v4_flash_eight_model_five_cell.json); only the DIPLOTYPE and DRUG fields are shown. Executed recommendations are computed by the authored mapping in _pgx_rules.py.